

Justin Hatch

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Anthropic Partner Program member via Machine & Minds. Currently shipping production LLM systems at Modern Amenities; previously built multi-agent trading infra (1st @ QuackHacks II) and ETL pipelines.

EDUCATION

University Of Oregon

B.S in Computer Science - Specialization in ML, AI & DS

Eugene, Oregon

Graduation: June 2026

- Dean's List | Department Honors | Oregon Blockchain Group | Oregon Software Consulting Group | QuackHacks II Winner

SKILLS

- Languages:** Python (Pandas, NumPy, Scikit-learn), SQL, C, C++, TypeScript, React, Node.js,
- Backend & Infrastructure:** REST API, Linux, Microservices, CI/CD, Pytest, FastAPI, Pydantic, Postgres,
- DevOps & Tools:** Streamlit, Docker, Airflow, AWS, Git, MLflow
- Machine Learning:** Feature Engineering, Predictive Modeling, XGBoost, PyTorch, HuggingFace Transformers, SHAP, Time Series Analysis, EDA, A/B Testing
- ML Engineering:** Model Evaluation, Cross-Validation, Hyperparameter Tuning, Model Deployment, Inference Optimization, Experiment Tracking, Feature Pipelines, Data Versioning
- AI Engineering:** Anthropic & OpenAI API, LangChain, Multi-Agent Systems, RAG, Vector Databases, Tool Use, Structured Outputs, Prompt Caching, LLM-as-judge Evaluation
- Certs:** [Data Camp - Associate Data Engineer in SQL](#)

PROJECTS

Market Data Pipeline + AI News Synthesis - Python | FastAPI | Supabase | AWS | Airflow | Alpaca | Yahoo Finance | LangChain

- Designed a Postgres medallion schema (Bronze → Silver → Gold → AI) keyed by (ticker, interval, timestamp), enabling replayable analytics across **4.5M+ rows** and unblocking downstream ML/agent workloads.
- Built replayable ETL ingesting **230+ tickers** across 1m-1d bars using conflict-aware upserts and historical backfills, achieving 100% idempotent reruns with zero duplicate rows across dozens of scheduled runs.
- Built a news ingestion pipeline (raw JSON → cleaned rows) and engineered **8+ technical features** (returns, SMA/EMA, Bollinger Bands, volatility) feeding the **AI synthesis layer** and downstream screening jobs.
- Deployed a scheduled **LangChain** agent that synthesizes market context, news, and financial reports into source-attributed summaries persisted to Postgres, powering retrieval for downstream RAG workflows.
- Containerized services and deployed to production on AWS App Runner; orchestrated Airflow DAGs with structured logging

NBA Salary Efficiency Analysis - Machine Learning | Sports Analytics | Residual Analysis | Python

- Built an XGBoost regression model on 1,500+ player-seasons to quantify NBA market efficiency, achieving $R^2 = 0.372$ with the unexplained variance treated as mispricing signal — surfacing the top 10 over/underpaid contracts league-wide.
- Tuned 700 hyperparameter configurations with 4-fold CV; compared error patterns and stability across folds.
- Performed residual analysis to surface “over/underpaid” signals and translate residuals into decision-ready insights.
- Scraped multi-year salary data from ESPN; cleaned, joined, and validated merges against player-season identifiers.
- Engineered features (OBPM, DBPM, VORP, position encoding, log transforms) to capture pay–performance relationships.

Evergreen Capital AI-Governed On-Chain Prediction Fund - Python | FastAPI | Gemini | TypeScript

- Architected a 5-agent Gemini research desk (Quant, Macro, Skeptic, Data Miner, Trader) where agents independently research, debate via structured cross-examination, and produce weighted consensus votes to trigger autonomous Polymarket trades.
- Built the FastAPI backend handling agent debate orchestration, weighted vote aggregation, and signal handoff to the on-chain execution layer, enabling autonomous trade triggers without human-in-the-loop.
- Built full-stack MVP in 24 hours with 2 teammates **1st Place Solana Track, 2nd Place Polymarket Track at QuackHacks II.**

Lethe - Multi-Agent LLM Consensus for Medical Bill Auditing - Python | FastAPI | OpenAI | Anthropic | Gemini | RAG | Blockchain

- Built a 3-agent LLM consensus system (OpenAI + Anthropic + Gemini) with role-specialized personas and calibrated confidence scoring, emitting streaming prose plus structured JSON votes over P2P 0G Mesh Network.
- Designed a round-2 reflection loop injecting peer findings as context with anti-herd-voting framing, plus a RAG priors loop feeding past audit outcomes into future runs so agents improve over time.

EXPERIENCE

Modern Amenities

Coburg, Oregon

AI Engineering Intern - Python | FastAPI | Claude | Prompt Caching | Prompt Engineering & Eval

March 2026 - Present

- Built FastAPI sales chatbot using Claude API qualifying prospects across **dozens of machine SKUs** in a 3D/AR demo serving **900+ app users**; used tool-use API to emit structured calls so frontend renders deterministic CTAs.
- Hardened the system prompt against 5 jailbreak vectors by wrapping untrusted input in delimiters and adding an output scanner, achieving a **100% block rate across all 5 tested attack patterns** with **zero system-prompt leaks**.
- Cut inference cost by implementing prompt caching with ephemeral **cache_control** on the static prefix and a 24h token budget, reducing input token costs by **upwards of 90%** on cached requests

Oregon Blockchain Group

Eugene, Oregon

Lead Software Engineer (promoted from Software Developer)

October 2025 – June 2026

- Led 2 engineering projects, **Crop Share** (farm to store supply-chain provenance) and **TrialWeave** (GLP-1 cohort RWE analytics) translating stakeholder needs into scoped specs and acceptance criteria, shipping across a team of **3-5 developers**.
- Drove 30%+ membership growth in under a year** by serving on the recruitment board, pitching in core CS classes, and interviewing potential candidates.